Trace synthesis: generate, verify, decontaminate

all counts measured on the released corpus - nothing projected

STAGE 1 · GENERATION

provider family A: generators

Expert labels



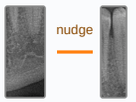
DENTEX · Tufts loaders
FDI conversion at ingest

ground truth per image
FDI 34 · Caries · bbox

Ground-truth-directed LangGraph loop

thought → tool call → observation

repeat every turn



before / after

generation pool:

- Gemini 3.5 Flash Lite
- MiniMax M3
- NVIDIA NIM
- local vLLM

tiered display perturbation, independent of the true box:
45% small · 25% big · 30% none - keeps accept-vs-nudge demonstrable

Unverified trace pool

```
T4 thought "crop shows only bone and
root area ... too low"
T4 call nudge_crop(dy=-0.30, s=0.60)
T5 obs bbox -> [1685, 885, 50, 193]
```

44,350 executed tool calls

2,292 corrective nudges

no training value until verified

STAGE 2 · VERIFICATION

re-verify until pass

provider family B - never the generator pool

Strict verifier

staged cross-family check per trace:

- ✓ diagnosis set-matches the full multi-finding ground truth
- ✓ evidence trail consistent - no claims on healthy tissue
- ✓ no directive leakage in CoT
- ✓ format + FDI notation valid

strict rate pacing via the provider pool

REJECT · 92

Repair loop

editor rewrites the trace,
keeping the valid CoT prefix:

- fix tool-arg / FDI slips
- drop phantom claims
- excise directive leakage

re-verified after every repair

logged

First-pass audit

92 rejections on 3,694 traces (97.5% pass)

hallucinated findings on healthy tissue — 66

contradicted the ground truth — 15

drift left uncorrected — 11

category 3 exists because perturbation pressure

PASS · 97.5% first-pass

PASS

STAGE 3 · DECONTAMINATION + RELEASE

only clean traces enter the corpus

Dual-gate decontamination

Gate 1 - regex sweep for directive phrasings leaked into the CoT

Gate 2 - flagged traces re-verified by family B before any action

105 traces excised

0 false positives on clinical "hint of"

clean

Released corpus

1,847 with-tools · 1,847 no-tools

exact per-cohort 1:1 parity (5 cohorts)

3,694 verified traces · 5,144 findings

687 healthy images · 44,350 tool calls

clean

Release

public on the Hugging Face Hub

per-call audit: perturbation tier, shown vs true bbox, offsets

every number regenerable

scripts/paper rebuilds all figures and tables in this paper